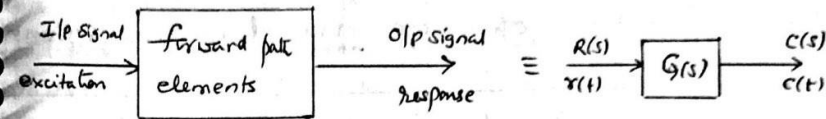


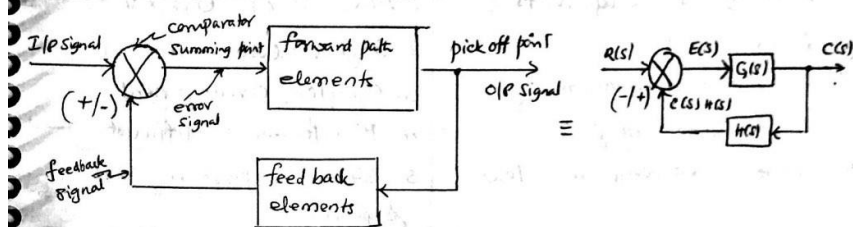
BLOCK DIAGRAMS

A block dia is a simple functional schematic that shows the various stages of a system by means of blocks along with their functions without indicating the type of components and their interconnections.

Block dia of an open loop system:



Block dia of closed loop system:



Transfer function: It is an input output relationship. It is defined as the ratio of L.T of O/P to the L.T of I/P.

$$T.F = \frac{C(s)}{R(s)}$$

open loop transfer function:

$$O/P = C(s) = G(s) \times R(s) \Rightarrow \frac{C(s)}{R(s)} = G(s) = \text{OLTF}$$

closed loop transfer function:

$$O/P = C(s) = E(s) \times G(s)$$

$$\text{Bw } E(s) = R(s) - C(s) \times H(s) \quad (\text{assuming -ve feed back}).$$

$$\therefore C(s) = [R(s) - C(s)H(s)]G(s)$$

$$= R(s)G(s) - C(s)G(s)H(s)$$

$$\therefore C(s)[1 + G(s)H(s)] = R(s)G(s)$$

$$\therefore \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} = \text{CLTF}$$

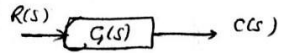
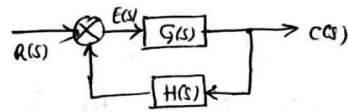
for a positive feedback,

$$CLTF = \frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)}$$

Note:

1. $G(s)$ is always referred as the open loop transfer function
2. $G(s)H(s)$ is referred as the loop transfer function
3. $1 + G(s)H(s) = 0$ is called the characteristic eqn of the system

Comparison between open loop and closed loop system:

Open loop System	Closed loop System
1. Less no of components 2. Lower cost 3. Simple arrangement 4. Maintenance easy 5. System response is slow 6. Stability uncertain 7. Poor efficiency	1. More no. of components 2. Higher cost 3. Complex arrangement 4. Maintenance difficult. 5. System response is quick & Superior. 6. System is more stable 7. More efficient system.
	

Significance of feedback: Feedback in a closed loop system allows comparison between the o/p and the i/p & thus makes the system performance better.

Advantages of feedback:

1. Improved system response
2. Improved quickness of desired response
3. Improved accuracy
4. Improves stability

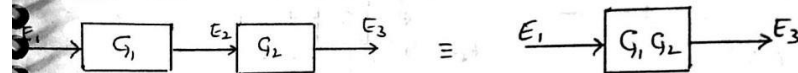
Note:

1. In some cases feed back causes the system to become unstable.
2. The feed back can be +ve or -ve (-ve is preferred)

Block diagram reduction Techniques:

Practical systems consist of complex block diagrams and multiple pick off points, summing points, feed backs. Hence in order to obtain a T.F., it is necessary to reduce the B.D to a simple form consisting of single loop. Thus the foll. techniques are used to reduce the block diagrams.

Blocks in cascade:

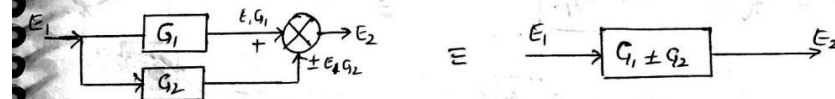


$$E_2 = E_1 G_1 \text{ \& } E_3 = E_2 G_2$$

$$E_3 = E_1 (G_1 G_2) \Rightarrow \frac{E_3}{E_1} = G_1 G_2$$

Thus the eq. of blocks connected in cascade is the product of block transfer functions.

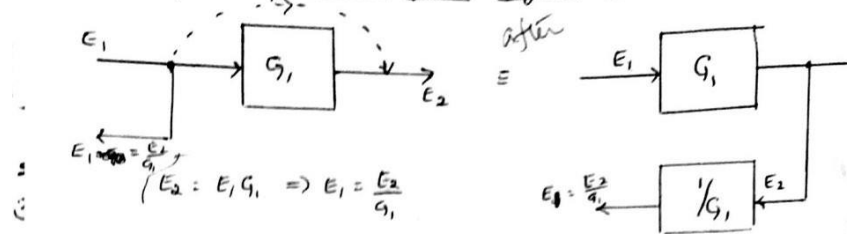
Blocks in parallel:



$$E_2 = E_1 G_1 \pm E_1 G_2 = E_1 (G_1 \pm G_2) \Rightarrow \frac{E_2}{E_1} = G_1 \pm G_2$$

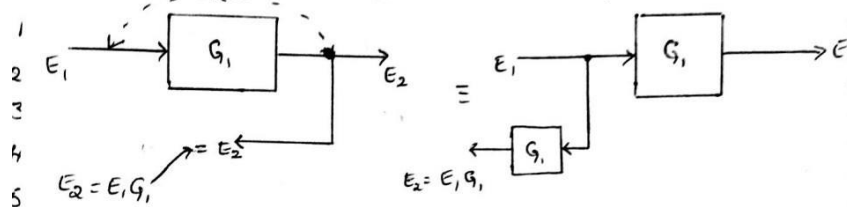
Thus, eq. of blocks connected in parallel is the sum (diff.) of two transfer functions of the blocks

3. Shifting the pick-off point beyond the block:



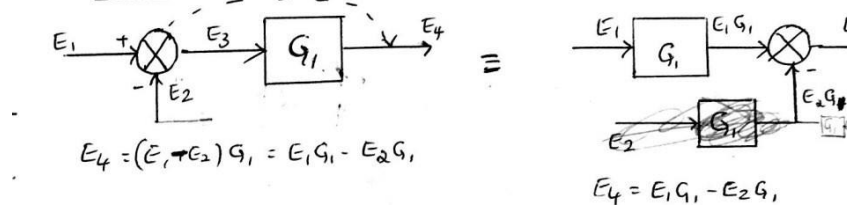
Whenever a pick-off point is shifted beyond the block, the picked off signal is to be divided by the block T.F.

4. Shifting a pick-off point ahead of the block:



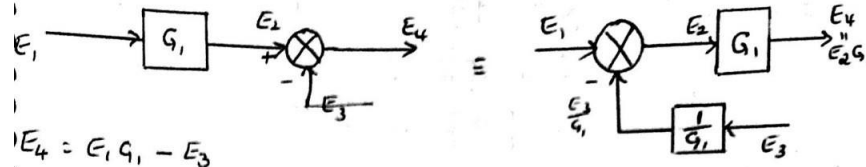
Whenever a pick-off point is shifted ahead of block, the picked off signal is to be multiplied by the block T.F.

5. Shifting the summing point beyond a block:



Whenever a summing point is shifted beyond the block the feed back signal (E_2) should be multiplied by the block transfer fn.

Shifting the summing point ahead of the block:



$$E_4 = E_1 G_1 - E_3$$

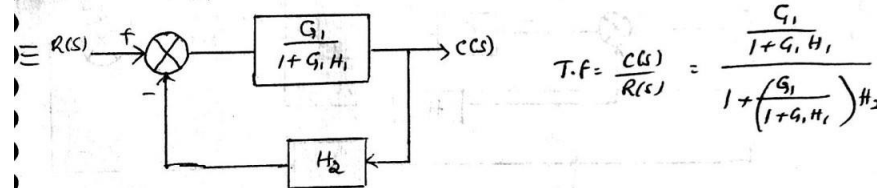
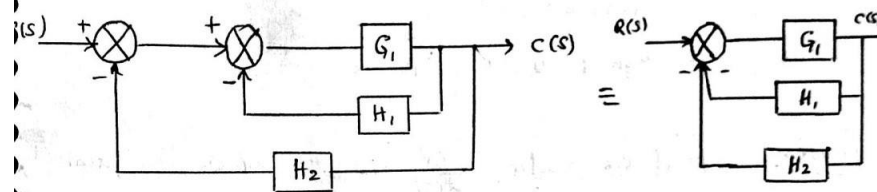
$$E_4 = E_2 G_1$$

$$E_2 = \left(E_1 - \frac{E_3}{G_1} \right)$$

$$\therefore E_4 = \left(E_1 - \frac{E_3}{G_1} \right) G_1 = E_1 G_1 - E_3$$

Whenever a summing point is shifted ahead of a block, the feed back signal has to be divided by the block transfer function.

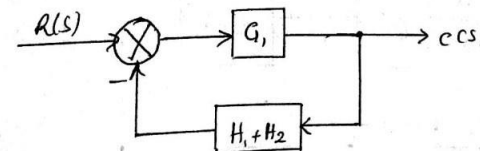
Feed backs in parallel: Consider a -ve feed back system



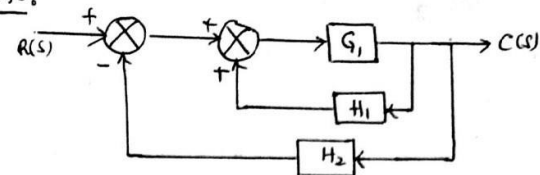
$$T.F = \frac{C(s)}{R(s)} = \frac{G_1}{1 + G_1 H_1 + \left(\frac{G_1}{1 + G_1 H_1} \right) H_2}$$

$$\therefore \frac{C(s)}{R(s)} = \frac{G_1}{1 + G_1 H_1 + G_1 H_2} \Rightarrow \boxed{\frac{G_1}{1 + G_1 (H_1 + H_2)} = T.F}$$

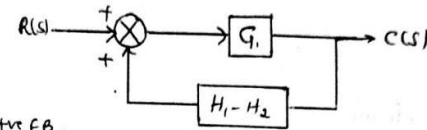
Thus the eq. of feed backs in parallel is equal to their algebraic sum. Thus the given B.D is eq. to



Note:



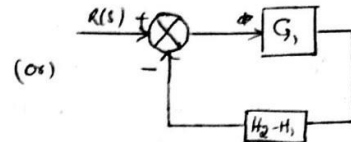
In this case either +ve FB or -ve FB can be chosen.



+ve FB.

$$T.F = \frac{C(s)}{R(s)} = \frac{G_1}{1 - G_1(H_1 - H_2)}$$

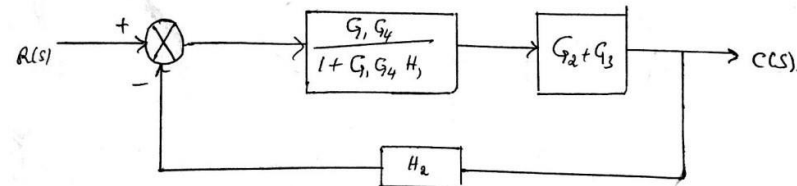
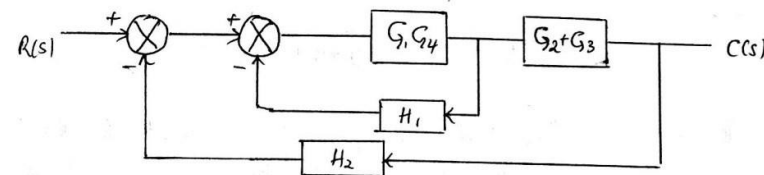
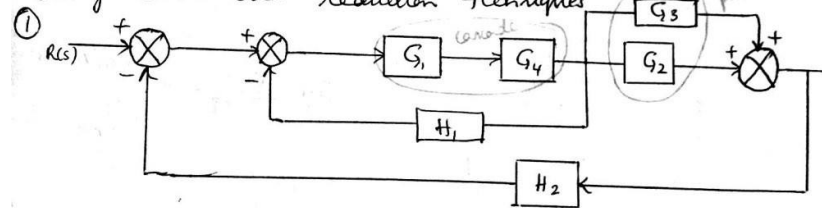
$$T.F = \frac{G_1}{1 + G_1(H_2 - H_1)} \quad (1)$$



$$T.F = \frac{C(s)}{R(s)} = \frac{G_1}{1 + G_1(H_2 - H_1)} \quad (2)$$

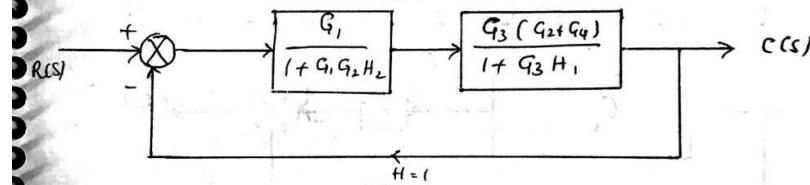
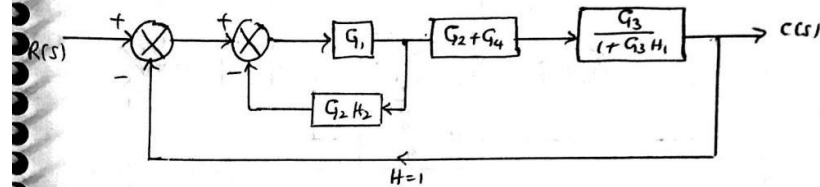
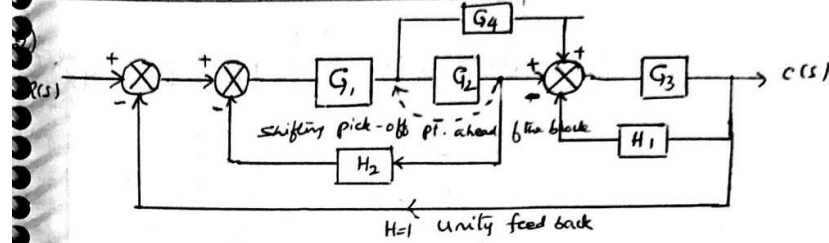
Thus both eqn (1) & (2) are same.

Problems: Find the ratio $\frac{C(s)}{R(s)}$ for the block diagrams given using block diagram reduction techniques.



$$\frac{C(s)}{R(s)} = \frac{(G_4 G_1 G_2 + G_3 G_4 G_1) / (1 + G_1 G_4 H_1)}{1 + \frac{[G_1 G_4 G_2 + G_3 G_4 G_1]}{[1 + G_1 G_4 H_1]} H_2}$$

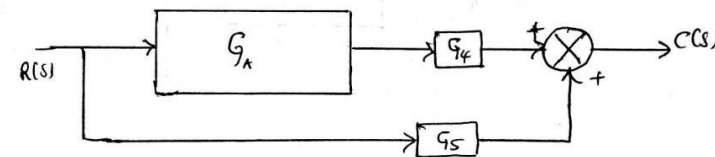
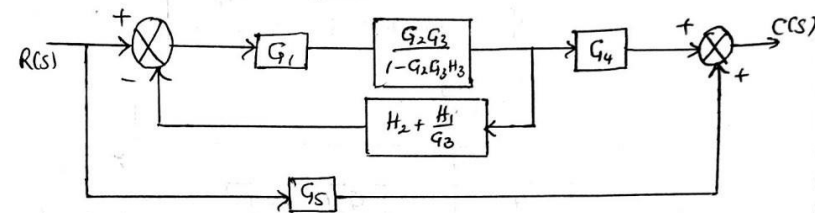
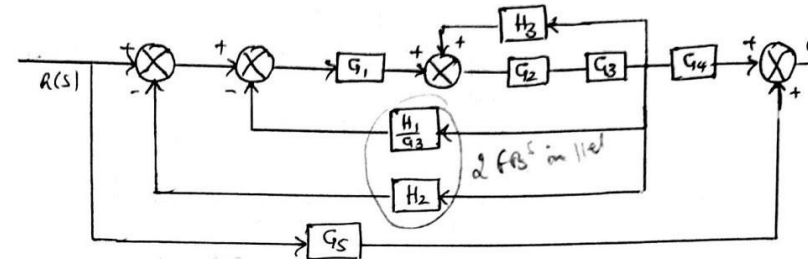
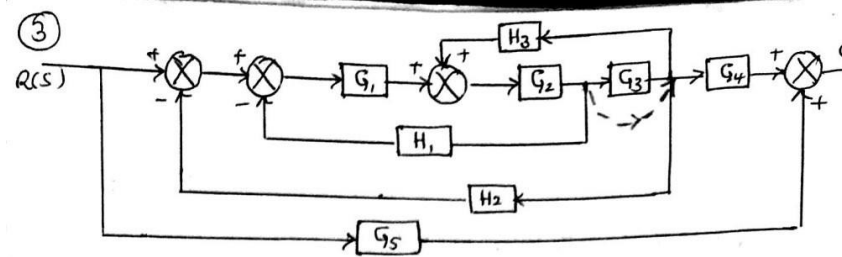
$$\frac{C(s)}{R(s)} = \frac{G_4 G_1 G_2 + G_3 G_4 G_1}{1 + G_1 G_4 H_1 + H_2 G_1 G_4 (G_2 + G_3)}$$



$$\frac{C(s)}{R(s)} = \frac{G_1 G_3 (G_2 + G_4)}{(1 + G_1 G_2 H_2)(1 + G_3 H_1)} \bigg/ 1 + \frac{G_1 G_3 (G_2 + G_4)}{(1 + G_1 G_2 H_2)(1 + G_3 H_1)} \times 1$$

$$\frac{C(s)}{R(s)} = \frac{G_1 G_3 (G_2 + G_4)}{(1 + G_1 G_2 H_2)(1 + G_3 H_1) + G_1 G_3 (G_2 + G_4)}$$

Note: if $H=1$ then T.F. = $\frac{G}{1+G}$

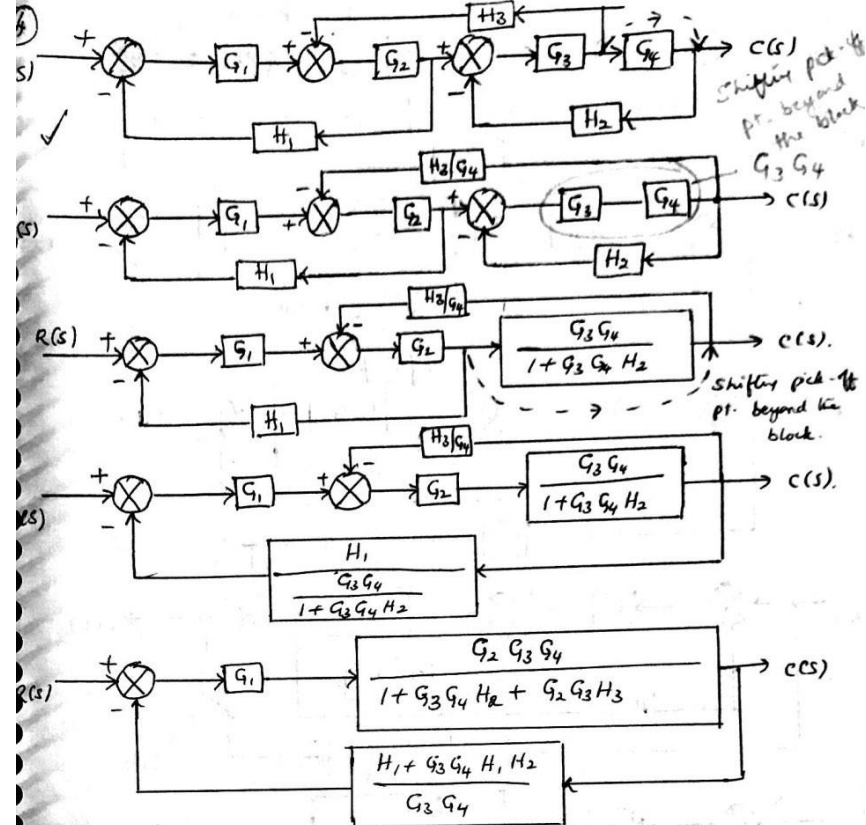


Where,

$$G_A = \frac{G_1 G_2 G_3}{1 + \frac{G_1 G_2 G_3}{1 - G_2 G_3 H_3} \left(H_2 + \frac{H_1}{G_3} \right)} = \frac{G_1 G_2 G_3}{1 + G_2 G_3 H_3 + G_1 G_2 G_3 H_2 + G_1 G_2 G_3 H_1}$$

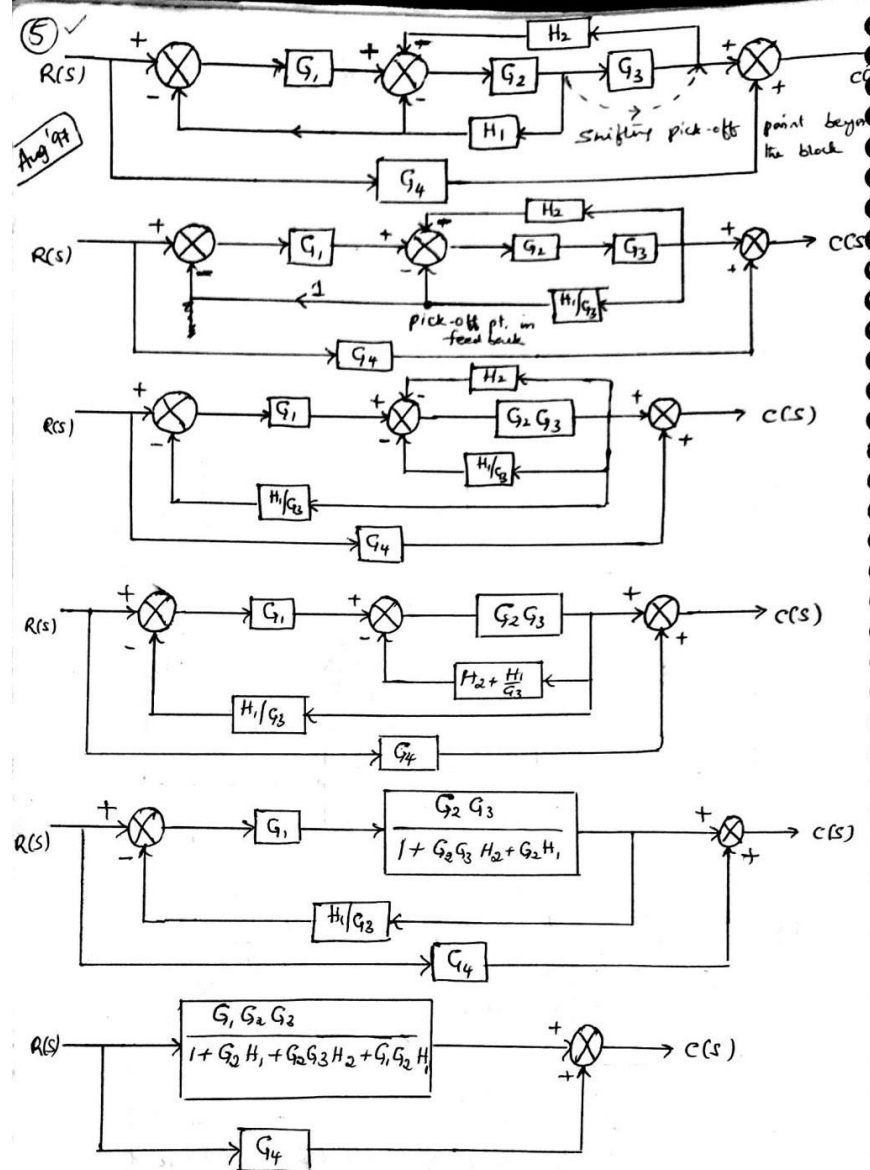
$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{1 - G_2 G_3 H_3 + G_1 G_2 G_3 H_2 + G_1 G_2 G_3 H_1} + G_5$$

$G_A G_4$ is in parallel with G_5 .

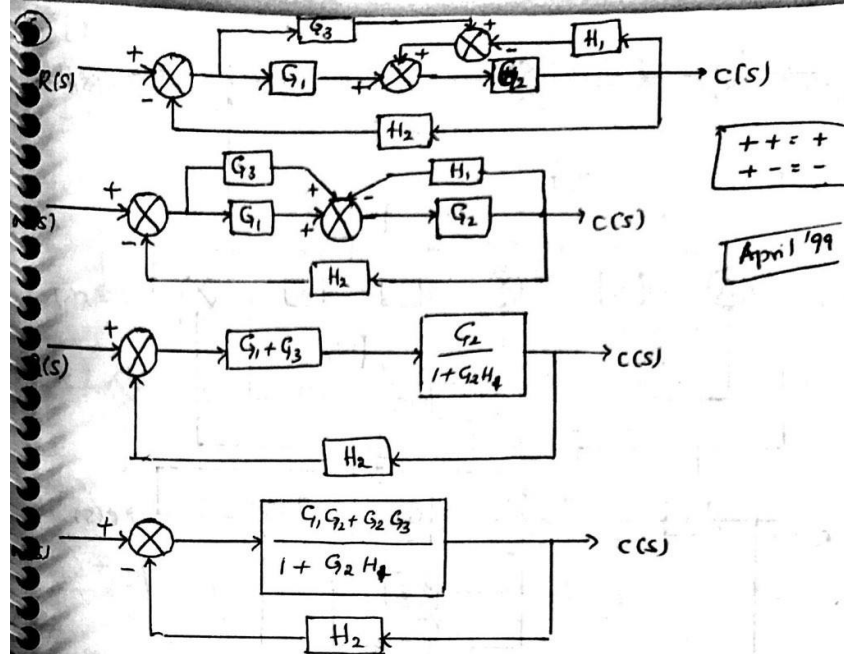


$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{1 + \left(\frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2 + G_1 G_2 H_2} \right) \left(\frac{H_1 + G_3 G_4 H_1 H_2}{G_3 G_4} \right)}$$

$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2 + G_1 G_2 H_2 + G_1 G_2 G_3 G_4 H_1 H_2}$$



$$\therefore \frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3}{1 + G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 H_1} + G_4$$

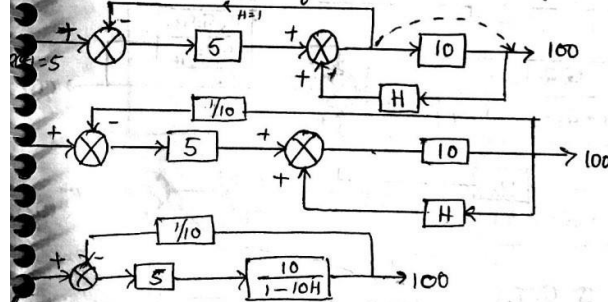


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April '99

$$T.F = \frac{C(s)}{R(s)} = \frac{G_1 G_2 + G_2 G_3}{1 + G_2 H_2 + G_1 G_2 H_2 + G_2 G_3 H_2}$$

In the B.D find the value of 'H'



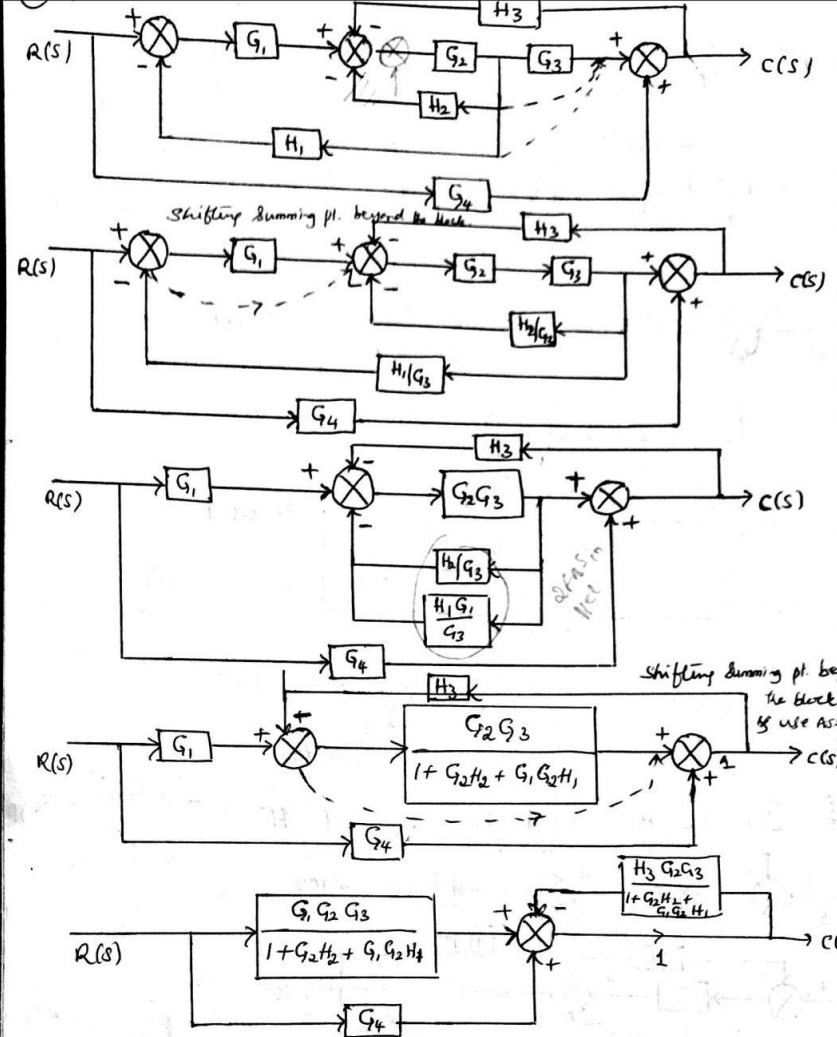
$$\frac{C(s)}{R(s)} = \frac{100}{5} = 20 = \frac{50}{1 + \frac{50}{1-10H} \times \frac{1}{10}} \Rightarrow 20 = \frac{50}{6-10H}$$

$$\therefore 6-10H = \frac{50}{20} = 2.5$$

$$10H = 3.5$$

$$H = 0.35$$

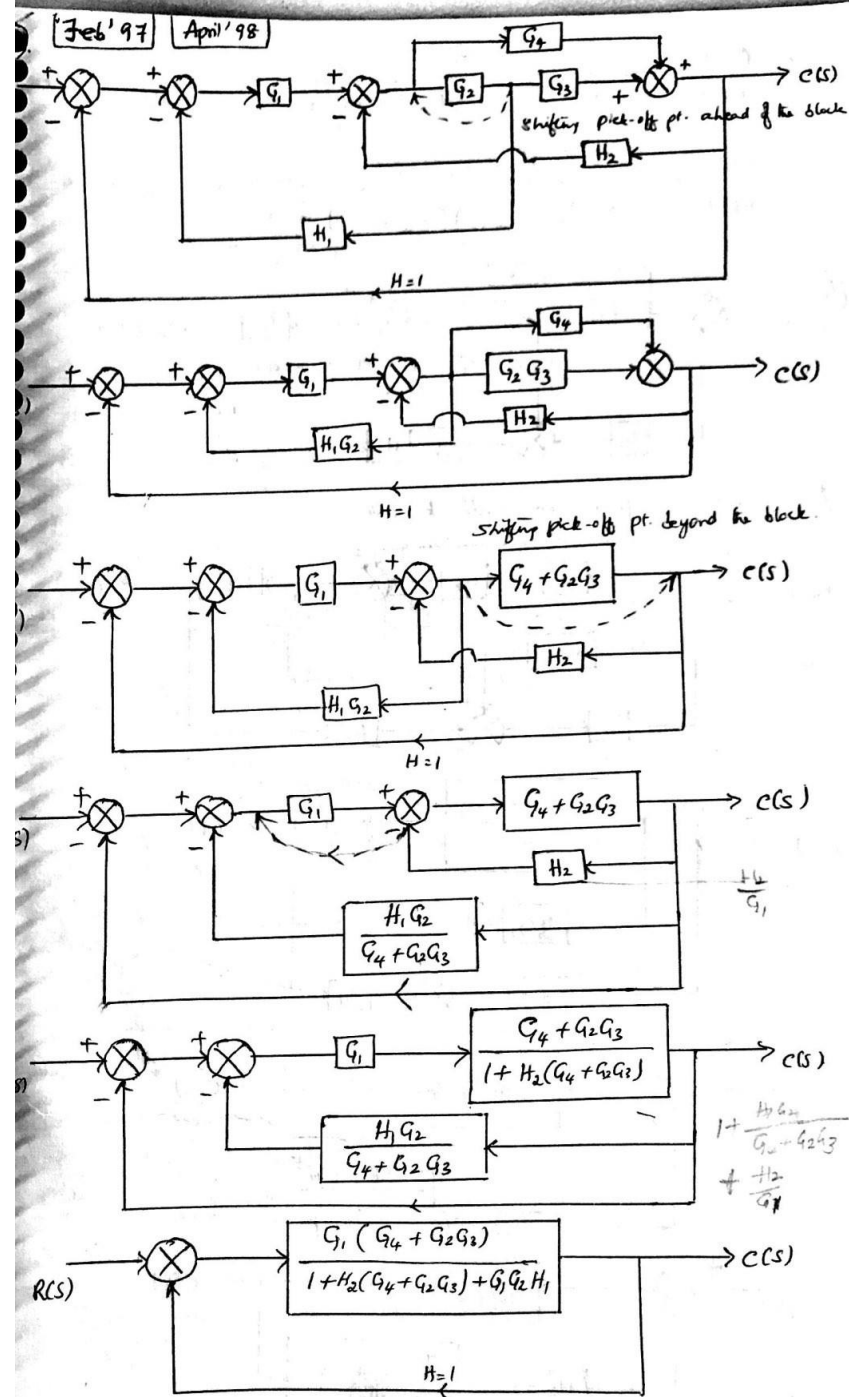
$$\frac{50}{1-10H+5}$$



$$\frac{C(s)}{R(s)} = \left[G_4 + \frac{G_1 G_2 G_3}{1 + G_2 H_2 + G_1 G_2 H_1} \right] \left[\frac{1}{1 + \frac{G_2 G_3 H_3}{1 + G_2 H_2 + G_1 G_2 H_1}} \right]$$

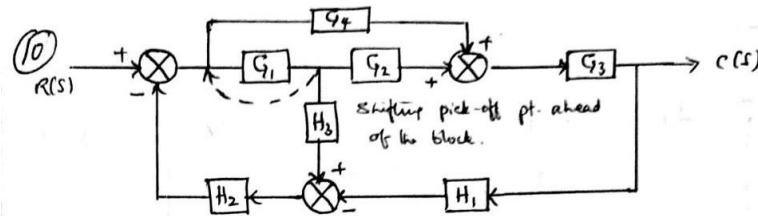
$$= \left(\frac{G_1 G_2 G_3 + G_4 + G_4 G_2 H_2 + G_1 G_2 G_4 H_1}{1 + G_2 H_2 + G_1 G_2 H_1} \right) \left(\frac{1 + G_2 H_2 + G_1 G_2 H_1}{1 + G_2 H_2 + G_1 G_2 H_1 + G_2 G_3 H_3} \right)$$

$$T.F = \frac{G_1 G_2 G_3 + G_4 G_2 H_2 + G_1 G_2 G_4 H_1 + G_4}{1 + G_2 H_2 + G_1 G_2 H_1 + G_2 G_3 H_3}$$

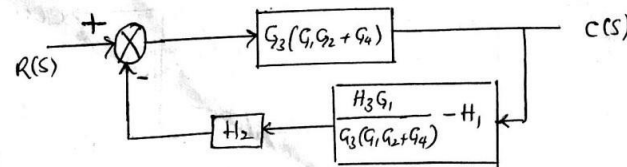
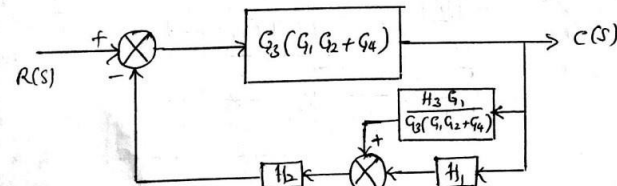
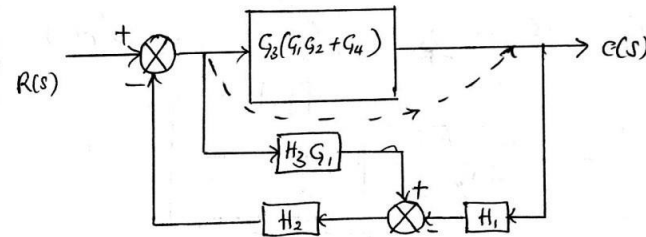
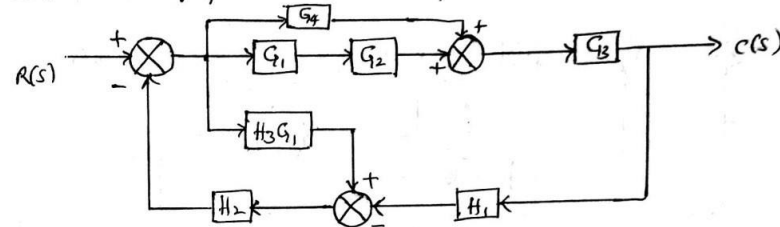


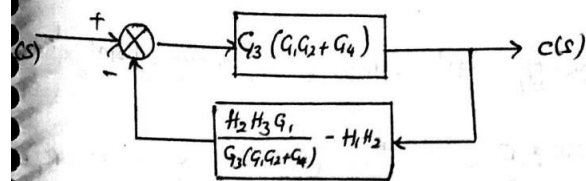
$$\therefore \frac{C(s)}{R(s)} = \frac{G_1 (G_4 + G_2 G_3)}{1 + (G_4 + G_2 G_3) H_2 + H_1 G_1 G_2 + G_1 (G_4 + G_2 G_3)}$$

$$\frac{C(s)}{R(s)} = \frac{G_1 (G_4 + G_2 G_3)}{1 + (G_4 + G_2 G_3) (H_2 + G_1) + H_1 G_1 G_2}$$



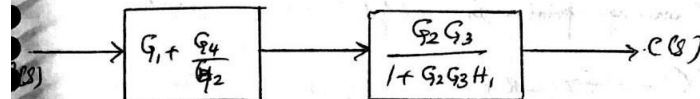
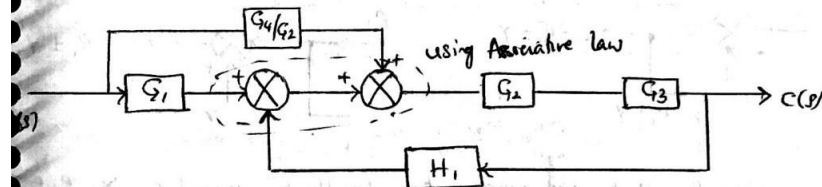
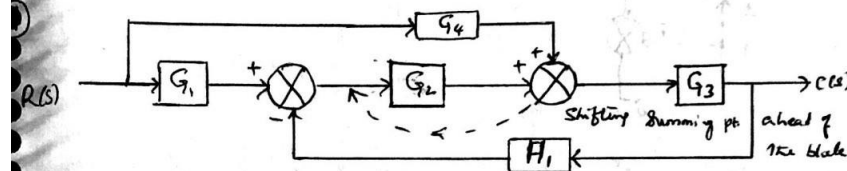
Note: Summing point in feedback path.





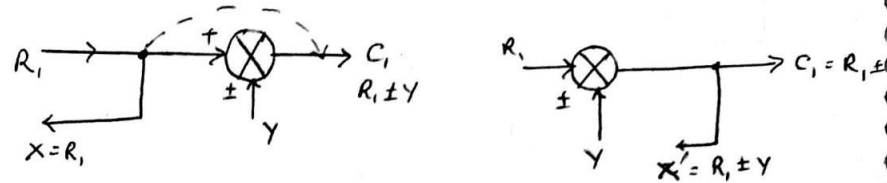
$$\frac{C(s)}{R(s)} = \frac{G_3(G_1G_2 + G_4)}{1 + G_3(G_1G_2 + G_4) \left\{ \frac{H_2H_3G_1 - H_1H_2G_3(G_1G_2 + G_4)}{G_3(G_1G_2 + G_4)} \right\}}$$

$$\frac{C(s)}{R(s)} = \frac{G_3(G_1G_2 + G_4)}{1 + G_1H_2H_3 - H_1H_2G_3(G_1G_2 + G_4)}$$

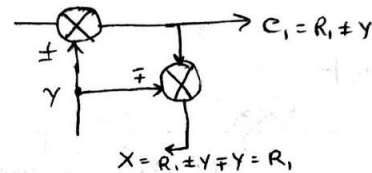


$$\frac{C(s)}{R(s)} = \frac{(G_1G_2 + G_4)G_3}{1 + G_2G_3H_1}$$

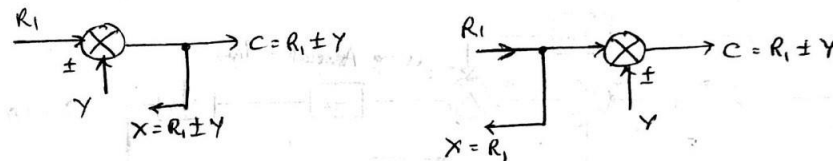
(1) Shifting take-off point after a summing point:



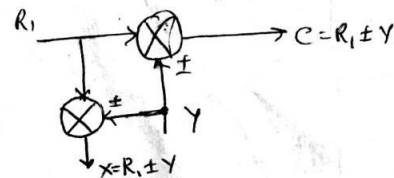
Here $X' = R_1 \pm Y$. But we want the ^{pickoff} feedback signal as $X = R_1$. So signal 'Y' must be inverted & added to so that X' becomes equal to R_1 . Thus, a summing pt. is reqd.

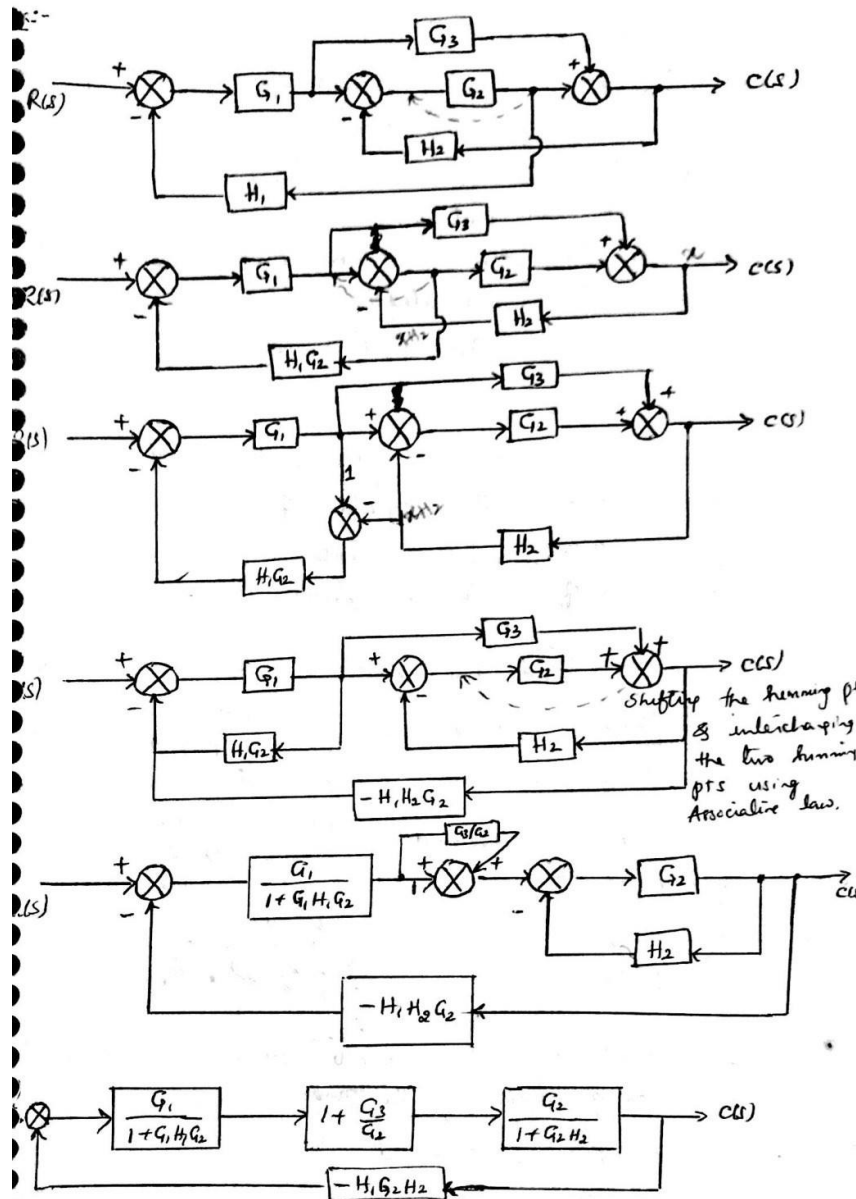


(2) Shifting take-off point after before a summing point:



Here $X = R_1$. But we want the feedback signal as $X = R_1 \pm Y$. Hence, signal Y must be added (with same sign) to R_1 . Thus a summing point is reqd.





On simplification,

$$\frac{C(s)}{R(s)} = \frac{G_1 (G_2 + G_3)}{1 + G_2 H_2 + H_1 G_1 G_2 - G_1 G_2 G_3 H_1 H_2}$$